

Transfer Learning for Dental Pathology Detection in Panoramic Radiographs: YOLOv11 vs. Zero-Shot Grounding DINO

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Abstract—Panoramic dental radiography is the most common supplementary imaging exam in dental practice, yet automated detection systems for it require annotated training data that is scarce and expensive to obtain. Zero-shot vision-language models offer an annotation-free path, but how accurately they localize pathologies on domain-specific images is not well characterized. This work compares Grounding DINO, a widely used open-vocabulary object detector, against a YOLOv11m model fine-tuned on 474 of the 678 annotated disease images from the DentexChallenge 2023 public dataset (950 after augmentation), targeting three pathologies: Caries, Periapical lesion, and Impacted tooth. On the DentexChallenge test split ($n = 103$), Grounding DINO achieved $\text{mAP}@50 = 0.00$ for all three classes across a range of prompt formulations, model sizes, and IoU thresholds down to 0.10, while the fine-tuned YOLOv11m reached $\text{mAP}@50 = 0.557$ ($\text{mAP}@50:95 = 0.361$), with Impacted tooth at 0.931 and Caries at 0.551. A three-stage pipeline was then applied to 50 externally sourced OPGs: YOLOv11 detection (466 total detections, mean 9.3 per image), spontaneous recall evaluation against dentist-written clinical descriptions (mean $\text{SR} = 89.7\%$ across 34 evaluable images), and LLM-based structured pre-clinical report generation via Gemini 2.5 Flash (BERTScore $\text{F1} = 0.779$ raw, -0.309 rescaled, on 50 reports). An input ablation isolating the contribution of the image versus the detection summary to report text found that neither input alone was clearly superior on text-similarity metrics, a result we attribute to the brevity of the reference descriptions rather than to the image being uninformative. The results show that the evaluated zero-shot detector (Grounding DINO Tiny) is insufficient for clinical OPG analysis at standard IoU thresholds, and that supervised fine-tuning on a small public dental dataset produces a usable detection baseline.

Index Terms—panoramic radiography, dental pathology detection, YOLOv11, Grounding DINO, zero-shot detection, transfer learning, DentexChallenge, automated report generation

I. INTRODUCTION

Panoramic dental radiography (orthopantomography, OPG) produces a single wide-field exposure covering the entire

dentition, mandible, maxilla, and surrounding structures. It is the standard first-line imaging request in dental clinical practice, used for screening caries, periapical lesions, bone loss, impacted teeth, retained roots, and orthodontic findings [1]. Systematic interpretation takes time and requires experience, and access to dental radiologists varies considerably across healthcare settings.

Automated detection systems based on convolutional neural networks and, more recently, transformer architectures have shown promising results on specific dental imaging tasks: Tuzoff et al. [2] applied Faster R-CNN to tooth detection and numbering in OPGs, Ekert et al. [3] used a U-Net for periapical lesion detection, and Alqhtani et al. [4] benchmarked multiple YOLO variants against 17 fine-grained dental anomalies, reporting $\text{mAP}@50$ between 0.54 and 0.78 depending on pathology. All these approaches depend on large, fully annotated bounding box datasets, which take considerable clinical expertise to produce.

A possible path around that bottleneck comes from vision-language models (VLMs) trained at internet scale. These models learn open-vocabulary visual representations that can be queried at inference time using text prompts, without any task-specific fine-tuning [5]. Grounding DINO [6], presented at ECCV 2024, extends the DINO transformer architecture with grounded pre-training and reported 52.5 AP on COCO zero-shot at publication. Deploying such a model directly on OPGs, prompted with disease names and without annotation effort, is an appealing idea.

This work tests whether that idea holds in practice. Specifically: at a standard IoU threshold of 0.5, does Grounding DINO detect Caries, Periapical lesion, and Impacted tooth at a useful level of precision? If not, what does supervised fine-tuning on the largest available public dental dataset achieve on the same task, and with what training resources?

Grounding DINO achieved $\text{mAP}@50 = 0.00$ on all three classes. YOLOv11m fine-tuned on 474 DentexChallenge training images (950 after augmentation) reached $\text{mAP}@50 = 0.557$ overall, with per-class results that reflect training data imbalance directly. A complete end-to-end pipeline then applied those detections to the 50 external OPGs and fed the output to a multimodal LLM to generate structured pre-clinical reports, evaluated against dentist-written reference descriptions using standard NLP metrics. Code and data splits are publicly available at <https://github.com/ErnaneJ/orthopantomography>.

II. RELATED WORK

A. Supervised Dental Pathology Detection

Convolutional detectors applied to OPGs generally report mAP in the 0.50–0.80 range for well-represented classes. Alqhtani et al. [4] found that performance varies sharply by pathology, with visually distinctive findings such as impactions and large restorations outperforming subtle ones such as early caries and periodontal bone loss by 20–30 mAP points within the same model, a per-class pattern that also appears in our results. The DentexChallenge 2023 [7], [8], a MICCAI workshop challenge, provided a public benchmark with disease-level bounding box annotations across three pathology classes, used here as both training source and evaluation benchmark.

B. Zero-Shot and Open-Vocabulary Detection

Grounding DINO [6] uses a Swin Transformer image encoder paired with a BERT-based text encoder, fused through a feature enhancer, to produce open-vocabulary bounding box predictions from text prompts. Prior work applying CLIP [5] to zero-shot classification in medical imaging found that general-purpose vision-language representations transfer to clinical domains for coarse classification, but noted performance gaps relative to supervised models [9]; localization at $\text{IoU} \geq 0.5$ is substantially harder than classification, and our experiments confirm a larger gap here.

C. Automated Radiology Report Generation

Large multimodal language models have been applied to chest radiograph reporting [10], with evaluation using BLEU [11], ROUGE-L [12], and BERTScore [13]. BERTScore F1, computed from contextual token embeddings, correlates more reliably with human quality ratings than n-gram metrics when reference and hypothesis texts differ substantially in length and style [13]. That is the case in our Stage 3 evaluation. Dasanayaka et al. [14] applied a multimodal LLM to OPG report generation and question answering; the 50 externally sourced OPGs evaluated in our Stage 3 (Section III-A) are drawn from the public Apache License 2.0 subset of their DPT Image and Caption Dataset [15]. Our task decomposition and evaluation protocol still differ from theirs (three separate stages of detection, recall, and report generation, evaluated against independently produced dentist descriptions, versus their single end-to-end captioning model evaluated against the dataset’s original captions), so results are not directly comparable, even though the source imagery overlaps. Balel

et al. [16] trained a detector on 30,954 panoramic radiographs, converted the output to structured JSON, and generated reports with several locally hosted and commercial LLMs evaluated by specialists; the commercial models tested, including a Gemini variant, hallucinated findings in every report reviewed. That result, at almost two orders of magnitude more training images than used here, cautions against treating LLM-generated dental reports as reliable without the specialist review step we identify as a limitation and future work below.

III. MATERIALS AND METHODS

Fig. 1 summarizes the three-stage pipeline described in this section: YOLOv11 detection (Stage 1, Section III-B), spontaneous recall evaluation against dentist descriptions (Stage 2, Section III-C), and LLM-based report generation (Stage 3, Section III-D).

A. Datasets

DentexChallenge 2023 (training and evaluation). The DentexChallenge 2023 quadrant-enumeration-disease subset [7], [8] is the largest publicly available OPG dataset with instance-level disease annotations (CC0 license). The full release contains 1,154 images; of these, 678 contain at least one bounding box annotation for the four disease categories used here: Caries, Deep Caries (merged with Caries in preprocessing), Periapical Lesion, and Impacted tooth. The remaining images have only quadrant or tooth-enumeration labels and were excluded from training and evaluation. The class distribution is skewed: across the 678 annotated images there are 2,767 Caries instances, 604 Impacted tooth instances, and 158 Periapical lesion instances (3,529 total). Images were split 70/15/15 into train ($n = 474$), validation ($n = 101$), and test ($n = 103$) sets using stratified random sampling (seed = 42). Per-split instance counts (Caries / Periapical lesion / Impacted tooth) are 1,934 / 111 / 458 in train, 408 / 25 / 77 in validation, and 425 / 22 / 69 in test. Images containing Periapical lesion or Impacted tooth annotations in the training split received two additional augmented copies each (Section III-B), yielding 950 effective training images and per-class training totals of 3,942 / 333 / 1,374 after augmentation.

External OPG dataset (pipeline evaluation). Fifty panoramic radiograph images (JPEG format, mean resolution approximately 2976×1488 pixels) were drawn from the publicly released subset of the DPT Image and Caption Dataset accompanying Dasanayaka et al. [14], [15], made available under the Apache License 2.0. The original authors report obtaining ethical clearance and informed consent for this data from Teaching Hospital Peradeniya and the Postgraduate Institute of Science, University of Peradeniya (Sri Lanka), with personally identifiable information removed prior to release (see Ethical Statement). Independent written clinical descriptions for these 50 images were subsequently produced by a trained dentist, co-author Y. González Sánchez, at the Fundación Odontológica Social Luis Seiquer (Seville, Spain); thirteen of these additionally have an audio recording of the verbal report. These independently produced

descriptions, not the original dataset captions, serve as the reference text for the Stage 2 and Stage 3 evaluations below. Bounding box annotations are not available for this dataset; it was used exclusively for the Stage 2 recall and Stage 3 report generation evaluations.

B. Stage 1: Detection Model

Preprocessing and augmentation. Images were resized to 640×640 pixels using letterbox padding. Training augmentation included CLAHE (clip limit 3.0) for X-ray contrast normalization, horizontal flip ($p = 0.5$), random brightness and contrast perturbation (± 0.2 , $p = 0.5$), Gaussian noise ($p = 0.3$), and affine transformations (scale 0.9–1.1, translate $\pm 5\%$, rotate ± 5 , $p = 0.4$), implemented with Albumentations. Color augmentations (HSV hue and saturation) were disabled because OPGs are grayscale. Images containing Periapical lesion or Impacted tooth annotations received two additional augmented copies each, tripling their effective training frequency.

YOLOv11m fine-tuning. YOLOv11m (medium variant, approximately 20M parameters) was initialized from COCO-pretrained weights and fine-tuned on the DentexChallenge training split using the Ultralytics framework [17]. Hyperparameters: AdamW optimizer, initial learning rate 10^{-3} , linear decay to 1% over training, batch size 8, image size 640×640 , up to 100 epochs with early stopping (patience = 20). Training ran on an Apple M5 MacBook Air (16 GB unified memory) using the Metal Performance Shaders backend. The model reached best validation mAP@50 = 0.547 at epoch 27 and stopped at epoch 41. Table I results (Section IV-A) follow the standard Ultralytics mAP evaluation protocol, which sweeps a low confidence threshold (≥ 0.001) to construct the full precision-recall curve; the operational threshold used for the qualitative Stage 1 output on the external OPGs (Section IV-B) and for the fixed-decision comparison against Grounding DINO in Fig. 2 is ≥ 0.25 , which trades recall for precision at deployment time. These are two different, individually standard use cases for the same trained model and should not be read as inconsistent thresholds for the same number.

YOLOv11n comparison. A smaller YOLOv11n variant (nano, 2.62M parameters, 181 layers, 6.6 GFLOPs) was trained under the same protocol for 50 epochs, reaching best validation mAP@50 = 0.484 at epoch 32. The gap to YOLOv11m’s 0.547 reflects the capacity difference between the two variants; YOLOv11m was used for all downstream pipeline stages, and YOLOv11n is reported here only as a lightweight-deployment reference point.

Grounding DINO zero-shot baseline. Grounding DINO Tiny (IDEA-Research/grounding-dino-tiny, Hugging Face Transformers) was evaluated on the same test split without fine-tuning. The three class names were used as text prompts. Detection thresholds: box confidence ≥ 0.25 , text score ≥ 0.20 . Predictions were matched to ground-truth boxes at IoU ≥ 0.5 , consistent with COCO evaluation. Section V-A reports a broader ablation over prompt formulations, model

size (Tiny vs. Base), and IoU threshold to test whether this single configuration understates the model’s true performance.

C. Stage 2: Spontaneous Recall Evaluation

The 50 external OPGs were processed with the fine-tuned YOLOv11m. For each image, the corresponding dentist description was parsed to identify mentions of the three YOLO-trained classes through substring matching. The Spontaneous Recall (SR) score for image i is:

$$SR(i) = \frac{|\hat{C}(i) \cap C^*(i)|}{|C^*(i)|} \quad (1)$$

where $\hat{C}(i)$ is the set of classes detected by YOLOv11 and $C^*(i)$ is the set of classes mentioned in the dentist description, both restricted to the three trained classes. The detector processes each image with a fixed confidence threshold of 0.25 and receives no information from the reference text. Images for which no YOLO-trained class appeared in the description were excluded from SR statistics (16 images), leaving 34 evaluable images.

Mentions of $C^*(i)$ are extracted from the free-text description by substring matching against a keyword-to-class alias dictionary, written in English to match the dentist descriptions. The dictionary has 31 distinct alias keys mapping to 24 clinical classes; 8 keys affect the SR metric reported here: *caries* and *carious* $\rightarrow$ Caries; *periapical* $\rightarrow$ Periapical lesion; *impacted* and *impaction* $\rightarrow$ Impacted tooth. A manual audit of the 20 images initially excluded found three misspellings the original dictionary missed, *carie*, *careis*, and *imapcted*, all present verbatim in the dentist’s free-text notes; these were added as additional aliases to Caries and Impacted tooth respectively, which is why the excluded count above is 16 rather than 20. The remaining 23 keys match disease and treatment terms outside the three-class detection scope and do not influence SR.

In qualitative examples, detections are color-coded by agreement with $C^*(i)$: green marks a class present in both $\hat{C}(i)$ and $C^*(i)$, and orange would mark a detected class absent from the description; not every example contains both colors.

D. Stage 3: Automated Report Generation

Structured pre-clinical reports were generated for all 50 external OPGs using Gemini 2.5 Flash via the OpenRouter API. Each request contained: (1) the OPG image in base64 encoding, and (2) a summary of detections from Stage 1 (class, confidence, and spatial location for the top 15 detections). The reference clinical description was not included in the prompt. The model was instructed to produce a five-section structured report using FDI tooth notation and to report only findings visible in the image. All requests used the OpenRouter-routed `google/gemini-2.5-flash` alias, queried in June and August 2026; OpenRouter does not expose a fixed dated snapshot for this alias, so results reflect whichever backend revision Google had live at request time, a reproducibility constraint of API-hosted models that we did not control.

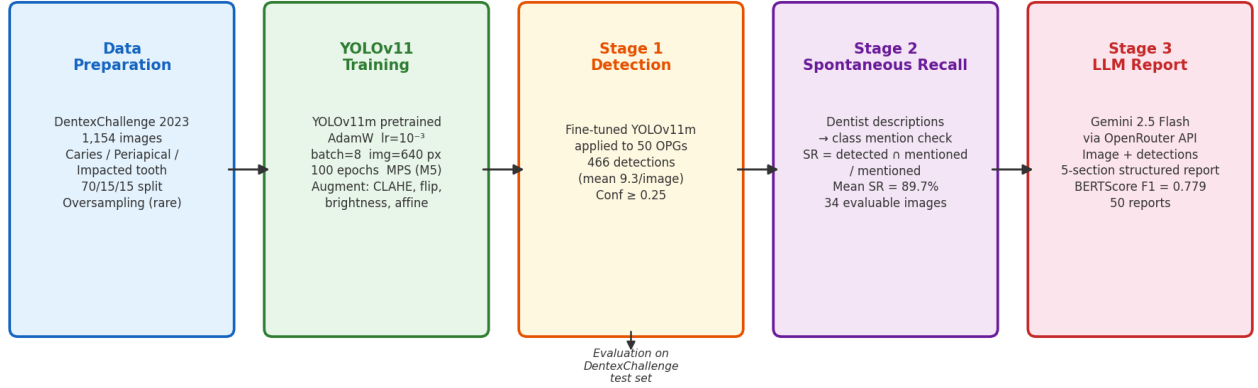


Fig. 1. Three-stage pipeline: YOLOv11 detection (Stage 1), spontaneous recall evaluation against dentist descriptions (Stage 2), and LLM-based report generation via Gemini 2.5 Flash (Stage 3).

Generated reports were compared to dentist-written reference descriptions using BLEU-4 [11], ROUGE-L [12], and BERTScore F1 [13] with RoBERTa-large embeddings.

To isolate the contribution of each input modality, the same 50 cases were regenerated under two additional conditions with the LLM, prompt structure, and evaluation held fixed: *image-only* (the OPG image with no detection summary) and *detections-only* (the Stage 1 detection summary as text with no image). Results are reported below (Section IV-D) alongside the default *both* configuration described above.

IV. RESULTS

A. Zero-Shot vs. Fine-Tuned Detection

Table I shows detection results on the DentexChallenge 2023 test split ($n = 103$ images), the only evaluation set with ground-truth bounding boxes.

TABLE I
DETECTION PERFORMANCE ON DENTEXCHALLENGE 2023 TEST SET (IoU = 0.5)

Model	Car.	Peri.	Imp.	mAP@50
GDINO-Tiny (zero-shot)	0.000	0.000	0.000	0.000
YOLOv11m (fine-tuned)	0.551	0.189	0.931	0.557

Car. = Caries; Peri. = Periapical; Imp. = Impacted. YOLOv11m evaluated at $\text{conf} \geq 0.001$ (standard Ultralytics protocol, full PR curve); GDINO-Tiny at $\text{conf} \geq 0.25$, $\text{text} \geq 0.20$ (Section III-B). Per-class mAP@50:95, P/R, and bootstrap CI reported in text below.

Grounding DINO produced bounding box outputs when queried with the three class names, but none reached $\text{IoU} \geq 0.5$ with any ground-truth box across the test split. The model generates text-anchored candidate regions that do not align closely enough with annotated pathology boundaries for clinical localization; Section V-A shows this holds across prompt formulations, model size, and looser IoU thresholds.

The fine-tuned YOLOv11m reached $\text{mAP@50} = 0.557$ (bootstrap 95% CI [0.501, 0.645], Section IV; mAP@50:95

$= 0.361$), with Precision = 0.583 and Recall = 0.550 at the confidence (≈ 0.30) that maximizes mean F1 across classes on the confidence sweep, not at the fixed $\text{conf} \geq 0.25$ deployment threshold used elsewhere in this paper. Fig. 2 shows the per-class comparison. Per-class results track the training data distribution: Impacted tooth ($\text{AP@50} = 0.931$) has a distinctive radiographic appearance and strong training coverage relative to its class size; Caries ($\text{AP@50} = 0.551$) is the most prevalent pathology but also the most morphologically variable; Periapical lesion ($\text{AP@50} = 0.189$) had only 158 annotated training instances.

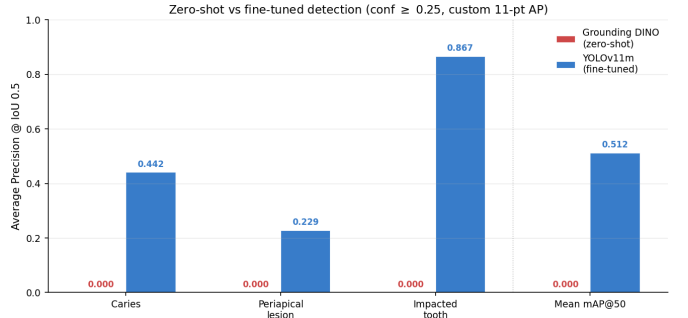


Fig. 2. Per-class AP@50 for Grounding DINO (zero-shot) and YOLOv11m (fine-tuned) on the DentexChallenge test set, both evaluated at the fixed $\text{conf} \geq 0.25$ operational decision threshold with a custom 11-point AP implementation (Section III-B); distinct from the confidence-sweep values in Table I.

B. Detection on the External OPGs (Stage 1)

Applied to the 50 external OPGs, the fine-tuned YOLOv11m produced 466 total detections (mean 9.3 ± 3.2 per image, range 3–20). By class: Caries 387 (83.0%), Impacted tooth 59 (12.7%), Periapical lesion 20 (4.3%). This distribution is close to the class balance of the training data itself (78.4% Caries, 17.1% Impacted tooth, 4.5% Periapical lesion of 3,529 annotated instances, Section III-A), which is the more direct explanation than a general claim about prevalence in unrelated

clinical populations; the low Periapical count also follows from the model’s weaker AP for that class.

Fig. 3 shows a representative Stage 1 output on one of the external OPGs. Multiple Caries detections appear bilaterally in the posterior sectors, a single Impacted tooth is flagged in the upper left third molar region, and a single Periapical lesion is flagged at low confidence (0.31) in an adjacent lower posterior region, consistent with that class’s overall lower AP@50. The color scheme follows the application convention: red for disease-category findings (Caries, Periapical lesion) and green for tooth-status findings (Impacted tooth).

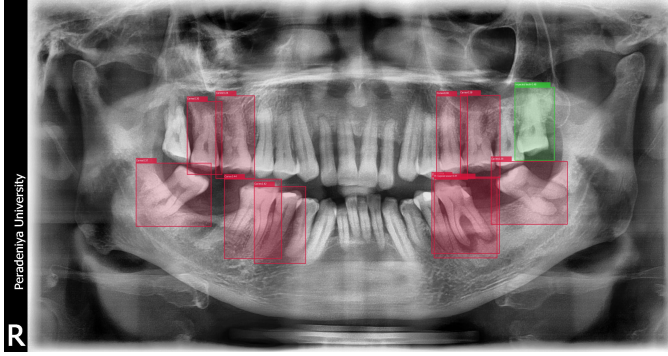


Fig. 3. Representative Stage 1 detection output on one of the external OPGs. Red bounding boxes indicate Caries and Periapical lesion detections (disease-category findings, bilateral posterior sectors); the green bounding box marks an Impacted tooth in the upper left third molar region. Confidence scores are displayed on each label; the Periapical lesion box has the lowest confidence (0.31), consistent with that class’s overall lower AP@50.

C. Spontaneous Recall (Stage 2)

Of the 50 external OPGs, 34 had dentist descriptions mentioning at least one YOLO-trained class. Mean SR across those 34 images was 89.7% (SD = 29.2%; median = 1.000). Table II shows the distribution.

TABLE II
STAGE 2 SPONTANEOUS RECALL DISTRIBUTION ($n = 34$ EVALUABLE IMAGES)

SR band	n	%
SR = 0.0	3	8.8
$0 < \text{SR} < 0.5$	0	0.0
$0.5 \leq \text{SR} < 0.8$	1	2.9
$\text{SR} \geq 0.8$	30	88.2
Wilson 95% CI for $\text{SR} \geq 0.8$: [73.4%, 95.3%].		

Two of the three images with SR = 0.0 (Image_23, Image_25) mention only Impacted tooth as the testable class, and YOLOv11m returned no Impacted tooth detection for either despite that class’s high aggregate AP@50 = 0.931 on DentexChallenge; the third (Image_48) mentions only Periapical lesion, consistent with that class’s low AP@50 = 0.189. The first two cases show that a class’s strong aggregate benchmark performance does not guarantee detection on every individual image. Given the small evaluable sample ($n = 34$), the Wilson 95% CI for the high-recall proportion is wide

([73.4%, 95.3%]); the point estimate of 88.2% should be read with that uncertainty in mind.

One case illustrates SR = 1.0 concretely: the dentist’s description mentioned a vertically impacted lower right third molar (tooth 48, FDI notation), and the model detected an Impacted tooth in the same image. YOLOv11m also detected Caries in this image, but Caries was not among the classes the dentist mentioned and is therefore not evaluated by the SR metric for this image.

D. Report Generation Quality (Stage 3)

Gemini 2.5 Flash generated structured five-section pre-clinical reports for all 50 images without error (136,257 input tokens, 26,057 output tokens). Table III shows NLP metrics against dentist-written reference descriptions.

TABLE III
NLP EVALUATION OF GENERATED REPORTS VS. DENTIST REFERENCE DESCRIPTIONS ($n = 50$)

Metric	Mean	SD	Embeddings
BLEU-4	0.0010	0.0008	n-gram
ROUGE-L	0.0315	0.0177	n-gram
BERTScore F1 (raw)	0.7792	0.0129	RoBERTa-large
BERTScore F1 (rescaled)	−0.3085	0.0767	RoBERTa-large

Rescaled BERTScore [13] is normalized against a general-domain baseline distribution; negative values reflect that structured, domain-specific report pairs sit outside that baseline’s typical range and are not by themselves evidence of poor semantic match (see text). Model: roberta-large, 17 layers, no IDF weighting (hash: roberta-large_L17_no-idf_version=0.3.12 (hug_trans=5.10.2)).

Low BLEU-4 and ROUGE-L values are expected: generated reports average approximately 300 words across five structured sections while dentist references average approximately 45 words of free text, so exact n-gram overlap is low regardless of semantic accuracy. Raw BERTScore F1 = 0.779 alone would suggest substantial shared semantic content, but this needs a control: to test whether BERTScore discriminates a correct report from an incorrect one here, rather than saturating near a floor for any English text pair in this domain, we recomputed it with each report paired against a mismatched reference description (a fixed derangement of the 50 cases, seed = 42, so no case is paired with its own reference). The mismatched-pair mean was 0.7777 raw (−0.317 rescaled) against 0.7792 raw (−0.309 rescaled) for correctly matched pairs; the two distributions overlap almost completely (SD $\approx$ 0.013–0.014 raw for both), so neither raw nor rescaled BERTScore separates correct from randomly wrong pairings on this sample. This confirms, rather than merely anticipates, the concern raised about this metric: neither variant is evidence of clinical accuracy, or even of same-versus-different-image correctness, for these short technical texts. The concrete example below carries more evidentiary weight than the aggregate BERTScore number.

A concrete example illustrates this gap: for Image_00, the generated report identified impacted teeth 18 and 38 with correct angulation descriptions, listed multiple carious lesions in FDI notation with severity estimates, and recommended

appropriate follow-up actions. The dentist’s reference note for the same image read: “Grossly carious 48 and deep caries in 36 with pulp exposure. Septic root of 25.” BLEU and ROUGE scores for this pair are near zero because the texts differ in structure and length; BERTScore F1 for this pair was 0.783 raw (-0.286 rescaled), reflecting that both texts describe the same pathological findings.

Table IV compares the default *both* configuration against the *image-only* and *detections-only* conditions described in Section III-D. Detections-only scored marginally higher than both other conditions on ROUGE-L and BERTScore F1 (raw and rescaled), while image-only scored lowest on every metric. BLEU-4 stayed near zero across all three and is not informative here for the reason given above. This pattern is discussed in Section V.

TABLE IV
INPUT ABLATION FOR STAGE 3 REPORT GENERATION ($n = 50$ PER CONDITION)

Condition	BLEU-4	ROUGE-L	BERT (raw)	BERT (resc.)
Image-only	0.0010	0.0304	0.7761	-0.3269
Detections-only	0.0015	0.0335	0.7956	-0.2113
Both (default)	0.0010	0.0315	0.7792	-0.3085

V. DISCUSSION

A. Localization Failure in Zero-Shot Detection

Grounding DINO is not blind to dental anatomy: when prompted with class names, it generates candidate boxes in plausible image regions. The failure is at localization precision, not semantic recognition. Standard mAP evaluation counts a detection as correct only if $\text{IoU}(\hat{b}, b^*) \geq 0.5$. For small lesions like periapical radiolucencies (typically a few millimeters in diameter at clinical scale), a box that roughly surrounds the affected region but is 30% too large fails this threshold entirely. Grounding DINO was trained to ground language to visual regions from natural image datasets, where objects are much larger and more salient than millimeter-scale dental pathologies. That spatial precision is simply not in what the model learned.

To test whether this failure is an artifact of prompt or model size, we ran the Tiny model under three prompt formulations (bare class names, descriptive phrases, full noun phrases) and added the Base model under the class-name prompt as a model-size check, each of these 4 prompt/model configurations evaluated at three IoU thresholds (0.50, 0.25, 0.10), for 12 conditions in total. mAP stayed at or near zero across all of them: the highest value observed at any condition, including $\text{IoU} \geq 0.10$, was 0.015 (Tiny, class-name prompt, Periapical lesion); most conditions scored exactly 0.000 even at $\text{IoU} \geq 0.10$, and Base scored no higher than 0.005 at any threshold, performing no better than Tiny. This rules out prompt engineering and model size as explanations: candidate regions do not consistently land inside the correct anatomical area at all, not merely outside the $\text{IoU} \geq 0.5$ boundary.

We also tested whether the fixed box confidence threshold (0.25) was filtering out correctly located but low-confidence boxes, re-running the Tiny model with the class-name prompt at confidence thresholds 0.05, 0.10, 0.15, and 0.25, crossed with the same three IoU thresholds. At the loosest IoU (≥ 0.10), lowering confidence from 0.25 to 0.05 nearly tripled mAP ($0.015 \rightarrow 0.041$); at the clinically relevant $\text{IoU} \geq 0.5$, mAP never exceeded 0.007 at any confidence tested. Confidence threshold matters only when the acceptance criterion already rewards approximately correct boxes; it does not rescue performance at the standard localization threshold, confirming the failure is in where boxes are placed, not how many are filtered before scoring.

In practice, this means that Grounding DINO, as evaluated here across both size variants and multiple prompt formulations, is not usable as a detection component in any pipeline that requires IoU-based localization on panoramic dental radiographs, even at thresholds well below the clinical standard. Whether other zero-shot or foundation detection architectures, or in-context/few-shot prompting strategies not tested here, would behave differently remains an open question. The result should be read as evidence against this specific family of models on this task, not as a general claim about zero-shot detection.

B. Transfer Learning from a Small Public Dataset

The $\text{mAP}@50 = 0.557$ from 474 training images (950 after augmentation) supports two conclusions. First, transfer from ImageNet/COCO pre-training generalizes to the X-ray domain at this dataset size when augmentation matches the modality (CLAHE, disabled color perturbation, oversampling of rare classes), consistent with recent evidence that YOLO-family transfer learning remains competitive with newer architectures on dental panoramic radiographs at small dataset scale [18]. Second, Impacted tooth at $\text{AP}@50 = 0.931$, a class among the most commonly missed findings in routine OPG review, is a promising starting point for computer-aided screening, pending clinical validation on larger and more diverse populations.

C. Spontaneous Recall as a Proxy Metric

The SR metric defined in (1) measures only whether the detector identifies at least one instance of each class the dentist mentioned: it does not measure spatial precision, does not penalize detections of classes not mentioned in the description (which may be genuine incidental findings or false positives), and does not validate bounding box accuracy. An image where the dentist wrote “caries present” and the detector produced ten spatially incorrect Caries boxes still scores $\text{SR} = 1.0$ for that class. SR is thus a class-level recall proxy, useful for characterizing pipeline behavior on the external OPGs where no bounding-box ground truth exists, but a necessary rather than sufficient condition for clinical utility; the high mean SR (89.7%) is consistent with the model’s detection rates on DentexChallenge and should not be read as a clinical accuracy figure.

D. Periapical Lesion: A Data Problem

Periapical lesion AP@50 = 0.189, well below Caries (0.551) and Impacted tooth (0.931), follows partly from having only 111 annotated training instances (333 after augmentation), a third of the next-smallest class; periapical radiolucencies are also small (2–10 mm), subtle in early stages, and easily confused with anatomical structures or noise, so some of the gap may reflect a harder visual task rather than data volume alone. This experiment cannot separate the two factors, since scaling annotated data while holding architecture fixed was not tested.

Asif and Khan [18] trained YOLO26l-seg on a Roboflow-processed DENTEX subset and reported mAP@50 = 0.591 across four disease classes, close to our 0.557 across three, but argue visual distinctiveness, not annotated volume, is the stronger per-class driver on this dataset. Our results do not settle this: Impacted tooth here has both the second-highest instance count and the most distinctive appearance, so volume and architecture are confounded in the same direction as in their setup. Separating them would require training on volume-matched subsets, which we did not do; we frame the data-volume account as a plausible partial explanation, not a demonstrated one.

Improving this class would require more training data, targeted augmentation, or semi-supervised learning from unlabeled OPGs, ideally evaluated against an architecture ablation to isolate how much of the gap is data-driven versus intrinsic to the detector.

E. Image and Detection Contributions to Report Text

Table IV shows that detections-only text matched the dentist’s reference description at least as well as, and on ROUGE-L and BERTScore slightly better than, the full image-plus-detections configuration used elsewhere in this paper; image-only scored lowest on every metric. Taken at face value this could be misread as evidence that the image contributes nothing once a detection summary is available, but we attribute it to a property of the reference texts rather than the image: dentist reference descriptions average approximately 45 words and list only the findings judged clinically salient, so a report instructed to describe only the detection list stays closer in content and length to that short reference. The full *both* condition gives the model room to describe additional visually plausible findings (e.g., restorations or edentulous spaces) outside the three trained classes and outside what the reference mentions, which lowers overlap with a short reference without indicating a worse report.

This does not show that the image is unnecessary. It shows that BLEU-4, ROUGE-L, and BERTScore, which all score similarity to a short reference text, cannot by themselves settle whether visual grounding improves report quality, since a terser, more constrained output tends to score better against a terse reference regardless of clinical completeness. A more informative test would compare the three conditions against dentist judgment of completeness and correctness directly, i.e., the clinical review step identified as a limitation above.

Practically, a text-only deployment tier, using the detection summary without re-transmitting the image to the LLM, is worth considering for cost and latency, provided a clinical reviewer confirms it does not omit visually apparent findings outside the detector’s three classes.

F. Web Prototype

To assess compatibility with clinical workflows, a proof-of-concept web application was built on the three-stage pipeline: a FastAPI backend runs Stage 1 detection via YOLOv11m, calls Stage 3 Gemini 2.5 Flash, and stores the annotated image and report; a React frontend overlays detection boxes with a raw/annotated toggle, lists findings by class, displays the five-section report in tabbed form, and accepts voice input for clinician corrections.

End-to-end latency from image upload to a fully rendered annotated radiograph and structured pre-clinical report was approximately 15 seconds on an Apple M5 MacBook Air with a standard broadband connection to the OpenRouter API. The dominant cost is the Gemini API call, not the local YOLO inference, which completes in under one second.

The prototype confirms that the computational requirements of the pipeline are compatible with commodity hardware: no discrete or datacenter GPU is required at inference time, and YOLO inference runs locally via Apple’s Metal Performance Shaders backend if the pipeline runs offline. This is a proof of concept for interactive deployment, not a clinically validated tool; a production system would require prospective clinical evaluation, regulatory compliance, and server-side inference at scale, outside the scope of this work.

G. Limitations

Four constraints limit these conclusions. First, quantitative detection evaluation relies on a single public benchmark (DentexChallenge 2023); performance on other equipment, populations, or annotation protocols may differ. Second, Stage 2 SR and Stage 3 NLP metrics are proxy evaluations on the external OPGs without bounding box ground truth: SR measures class-level agreement, not spatial precision or false-positive rate; BERTScore measures text similarity, not clinical correctness, and our random-pairing control (Section IV-D) shows it barely distinguishes a correct reference from a mismatched one here, so it carries little independent weight. The same limitation applies to the image-vs-detections ablation (Section IV-D): it measures similarity to a short reference, not clinical completeness, so it cannot alone determine whether the image is dispensable. Third, the generated reports were not reviewed by clinicians; a blinded expert evaluation is planned as immediate future work before any clinical utility claim can be made. Fourth, training and inference ran on an Apple M5 MPS backend, consumer hardware; throughput figures are a proof-of-concept reference, not a production benchmark.

VI. CONCLUSION

Grounding DINO, tested without fine-tuning across three prompt formulations and two model sizes, achieved mAP@50

= 0.00 (and near-zero even at $\text{IoU} \geq 0.10$) on the Dentex-Challenge 2023 test set for all three target classes because its localization precision is insufficient for millimeter-scale dental pathologies. YOLOv11m fine-tuned on 474 annotated training images (950 after augmentation) from the same public dataset reached $\text{mAP@50} = 0.557$ overall (95% CI [0.501, 0.645]), with Impacted tooth at 0.931, Caries at 0.551, and Periapical lesion at 0.189. Applied to 50 externally sourced OPGs without bounding box ground truth, the pipeline produced a mean spontaneous recall of 89.7% (class-level agreement with dentist descriptions, Wilson 95% CI [73.4%, 95.3%] for the high-recall proportion) and structured pre-clinical reports with BERTScore F1 = 0.779 raw (−0.309 rescaled; semantic text similarity against reference notes, though a random-pairing control found this metric barely separates correct from mismatched report-reference pairs on this sample); all figures are exploratory proxies pending clinical validation.

Grounding DINO, evaluated zero-shot across both size variants and multiple prompts, was not suitable for clinical dental detection here, even at IoU thresholds well below the clinical standard; this should not be generalized to all zero-shot or foundation models without further evaluation. Supervised fine-tuning on a small public dataset, with augmentation tailored to the modality, produces a viable detection baseline. A React/FastAPI web prototype of the complete pipeline delivers a structured pre-clinical report from image upload in approximately 15 seconds on commodity hardware, with no discrete or datacenter GPU required at inference time.

Future work will focus on scaling the annotated training set, exploring semi-supervised approaches for Periapical lesion, extending detection to the full 31-class clinical vocabulary, and obtaining clinical validation of the generated reports by specialist dentists.

ETHICAL STATEMENT

The DentexChallenge 2023 benchmark used for Stage 1 training and evaluation is released under a CC0 license and contains no identifiable patient information. The 50 external OPGs used for the Stage 2 and Stage 3 evaluations are drawn from the publicly available subset of the DPT Image and Caption Dataset [15] accompanying Dasanayaka et al. [14]. The original authors report that ethical clearance for this data was obtained from the relevant hospital ethics committees under international ethical standards, that informed consent was obtained from all subjects, and that the released images were de-identified prior to publication; the dataset is distributed under the Apache License 2.0. No additional patient contact, data collection, or identifiable information handling was performed by the present authors; our use is limited to secondary analysis of this already-published, de-identified, licensed dataset, together with independently authored clinical descriptions (see Section III-A). The authors declare no conflicts of interest.

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